INCH-POUND

MIL-DTL-12632/2B
25 April 2016
SUPERSEDING
MIL-L-12632/2A
10 June 1974

DETAIL SPECIFICATION SHEET

LOUDSPEAKER, PERMANENT MAGNET (ENCASED, 2-WATT, FUNGUS-, GUNBLAST-, AND IMMERSION-RESISTANT), TYPE LS-166/U

Inactive for new design after 30 March 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the loudspeaker described herein shall consist of this document and the latest issue of Specification MIL-DTL-12632.

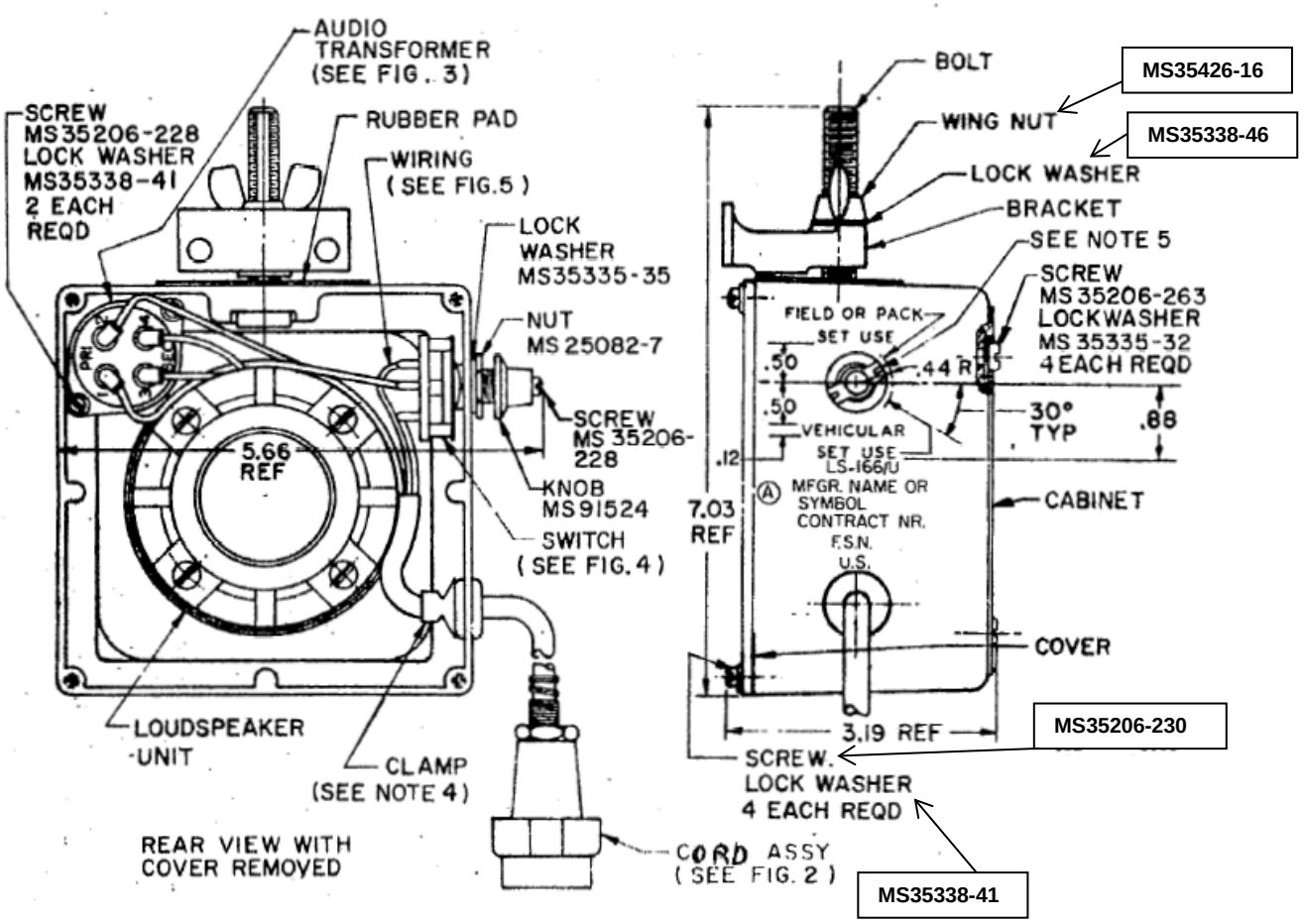


FIGURE 1. Loudspeaker LS-166/U.

AMSC

FSC 5965

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:07Z
Check the source to verify that this is the current version before use.

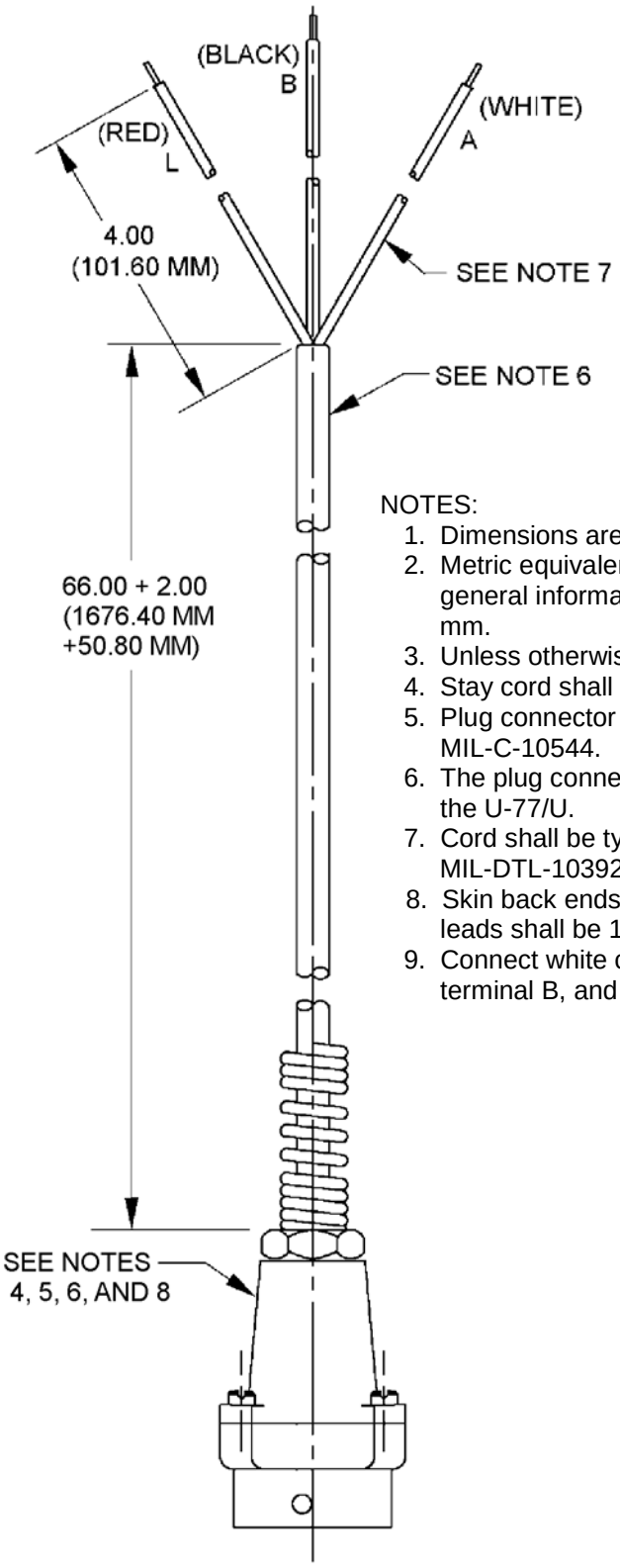
Component name	Applicable document	Part number
Nut	MS25082	MS25082-7
Screw	AIA/NAS NASM35206	MS35206-228, -230, -263
Lock Washer	AIA/NAS NASM35335	MS35335-32, 35
Lock Washer	AIA/NAS NASM35338	MS35338-41, 46
Wing nut	MS35426	MS35426-16
Knob	AIA/NAS NASM91524	MS91524

INCHES	MM
.12	3.05
.44	11.18
.50	12.70
.88	22.35
3.19	81.03
5.66	143.76
7.03	178.56

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ± .02 (.51 mm) and ± 1/2° on angles.
4. Steel clamp firmly clamped on cable jacket. Design is optional. A strain relief bushing (e.g., Heyco) is a suitable substitute for the grommet/clamp combination.
5. Drill .391 (9.93 mm) diameter hole for rotary switch.

FIGURE 1. Loudspeaker LS-166/U - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:07Z
Check the source to verify that this is the current version before use.

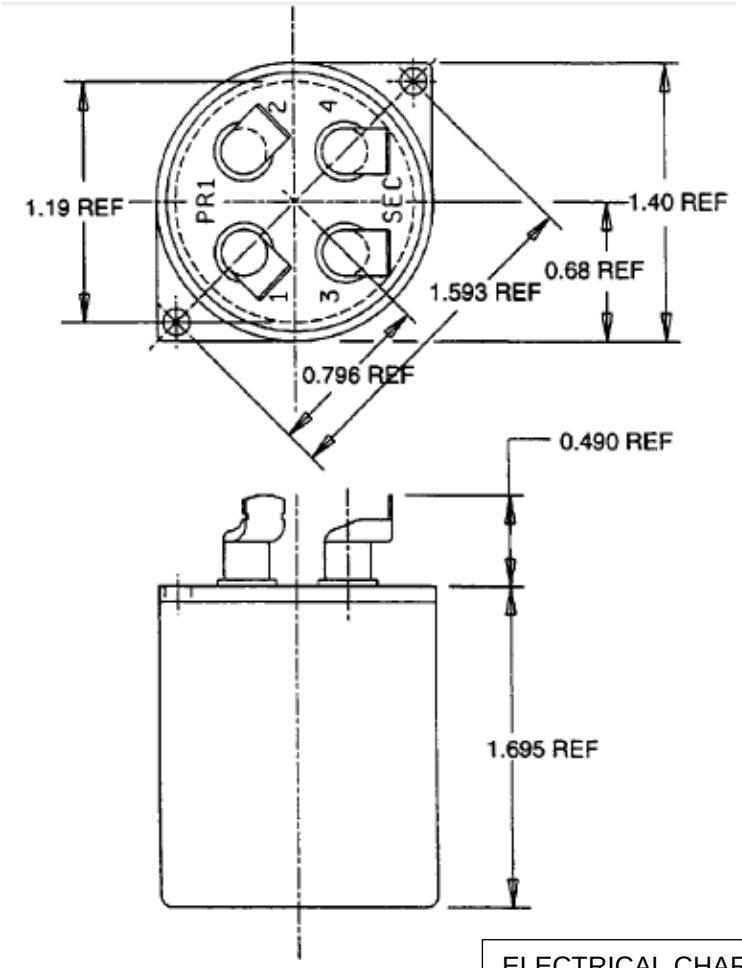


NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ± .06 (1.52 mm)
4. Stay cord shall be tied securely to stay ring of plug.
5. Plug connector shall be type U-77/U in accordance with MIL-C-10544.
6. The plug connector type U-161/U is a suitable substitute for the U-77/U.
7. Cord shall be type WT-15A/U in accordance with MIL-DTL-10392.
8. Skin back ends .12 (3.05 mm), twist and solder coat. Plug end leads shall be 1.12 (28.45 mm) long before assembly.
9. Connect white conductor to terminal A, black conductor to terminal B, and red conductor to terminal L of plug U-77/U.

FIGURE 2. Cord assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:07Z
Check the source to verify that this is the current version before use.



INCHES	MM
0.490	12.45
0.680	17.27
0.796	20.22
1.190	30.23
1.400	35.56
1.593	40.46
1.695	43.05

ELECTRICAL CHARACTERISTICS
Rated power level: 2 watts, ± 1 db
Frequency response range 250 to 5,000 Hz at rated power,
flat within 2 db.
Reference frequency 1,000 Hz
Primary impedance 600 ± 60 Ω
Secondary impedance 8 Ω
Insertion loss 0.5 db, max, with 2 watts input at reference frequency
Temperature rise 40°C, max, above an ambient of 65°C, with rated power at 300 Hz
Test voltage 500 volts, rms
Duty cycle Continuous
Maximum altitude 10,000 feet
Harmonic distortion Not more than 1 percent at rated power
Polarity Additive, with terminals 2 and 3 connected

FIGURE 3. Audio transformer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:07Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ± .02 (.51 mm).
4. Solder lug terminals, No. 18 AWG.
5. Markings shall be as shown.
6. Complete dimensions and configuration characteristics are in MIL-PRF-27/2, (Part number M27/2-01) Type TF4R21YY for a GRADE 4, CLASS R, FAMILY 21 type transformer.

FIGURE 3. Audio transformer - Continued.

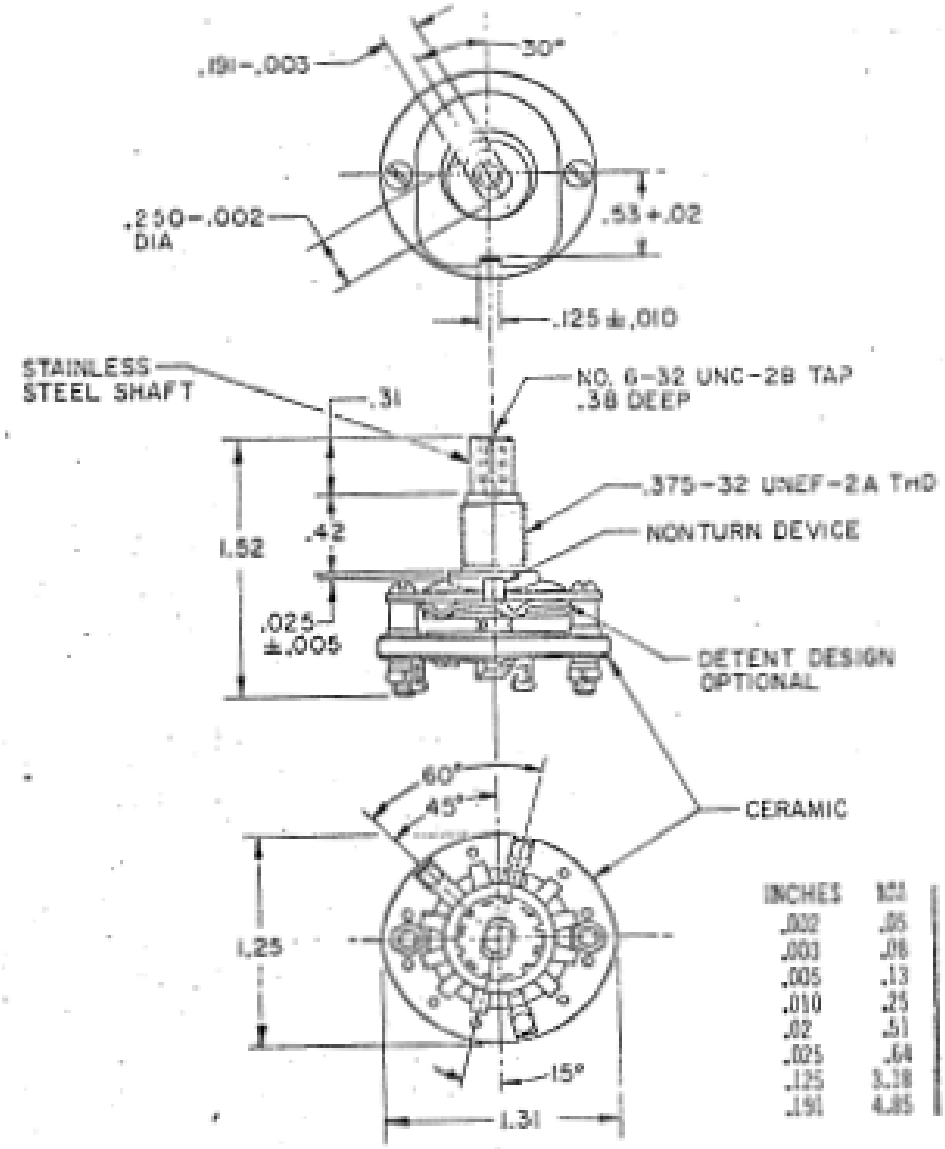


FIGURE 4. Rotary switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:07Z
Check the source to verify that this is the current version before use.

INCHES	MM	INCHES	MM
.002	.05	.250	6.35
.003	.08	.31	7.87
.005	.13	.38	9.65
.010	.25	.42	10.67
.02	.51	.53	13.46
.025	.64	1.25	31.75
.125	3.18	1.31	33.27
.191	4.85	1.52	38.61

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ± .02 (.51 mm) and ±1/2°.
4. Construction – single section, single pole.
5. Contacts shall be non-shorting.
6. Stops shall be provided to limit rotation to two positions.
7. Terminals – angle of bend 60° to 90°.
8. Torque (rotational) At 25° +10° -5°C: 1.5 pound-inches min: 6 pound inches max. at -55° ± 2°C: 1.5 pound-inches min: 8 pound inches max.
9. Life, rotational 10,000 rotational cycles at ± 125°C.

FIGURE 4. Rotary switch - Continued.

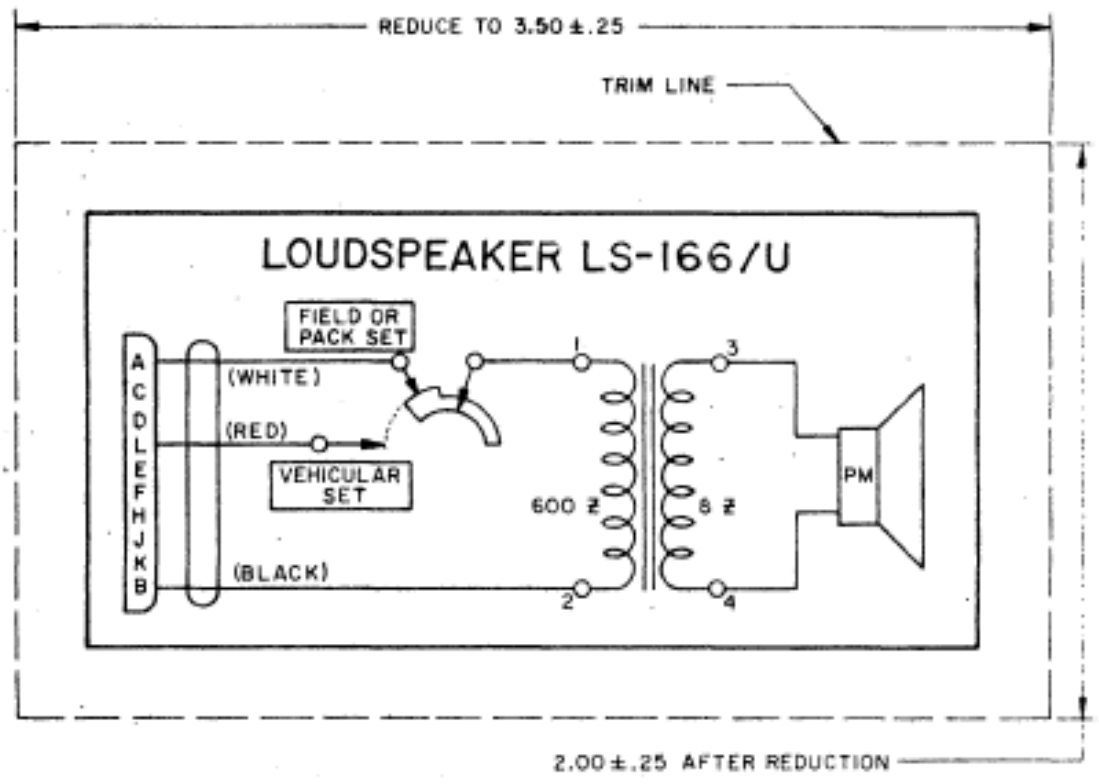


FIGURE 5. Schematic diagram and label size.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:07Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction (see figure 1)
Loudspeaker unit-Type LS-445/U of MIL-DTL-12606.
Cabinet-In accordance with MIL-DTL-12632.
Cord assembly-See figure 2.
Cable-Type WT-15A/U of MIL-DTL-10392.
Plug-Type U-77/U of MIL-C-10544.
Audio transformer-See figure 3.
Rotary switch-See figure 4.
Wiring-See figure 5.

Schematic-diagram label: In accordance with MIL-DTL-12632 and figure 5.

Frequency response test voltage (rms). 36.0 volts.

Marking See figure 1

The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based upon the entire content, regardless of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-DTL-12632, this document references the following:

AIA/NAS NASM35206
AIA/NAS NASM35335
AIA/NAS NASM35338
AIA/NAS NASM91524
MIL-DTL-10392
MIL-C-10544
MIL-DTL-12606
MIL-PRF-27/2
MS25082
MS35426

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85:
DLA – CC

Preparing activity:
DLA – CC

(Project 5965-2016-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:07Z
Check the source to verify that this is the current version before use.